

Probability & Statistics – One Page Formula Revision

Permutation

$$nPr = n! / (n-r)! \quad (\text{Order matters})$$

Combination

$$nCr = n! / [r!(n-r)!] \quad (\text{Order does not matter})$$

Probability

$$P(A) = \text{favourable outcomes} / \text{total outcomes}$$

$$P(A^c) = 1 - P(A)$$

Conditional Probability

$$P(A|B) = P(A \cap B) / P(B)$$

Bayes' Theorem

$$P(A|B) = [P(B|A) P(A)] / P(B), \quad P(B) = \sum P(B|A_i)P(A_i)$$

Naïve Bayes (Concept)

$$P(Y|X_1 \dots X_n) \propto P(Y) \prod P(X_i|Y)$$

Assumption: conditional independence

Expectation (Mean)

$$E[X] = \sum x P(X = x)$$

Variance & Standard Deviation

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{SD}(X) = \sqrt{\text{Var}(X)}$$

Covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Cov > 0: move together, Cov < 0: opposite

Linear Transformation

$$\text{If } Y = aX + b:$$

$$E[Y] = aE[X] + b$$

$$\text{Var}(Y) = a^2\text{Var}(X)$$

Bernoulli Distribution

$$X \in \{0, 1\}$$

$$P(X=1)=p, P(X=0)=1-p$$

$$\text{Mean} = p, \text{Var} = p(1-p)$$

Binomial Distribution

$$P(X=k) = nCk p^k (1-p)^{(n-k)}$$

$$\text{Mean} = np, \text{Var} = np(1-p)$$

Poisson Distribution

$$P(X=k) = (\lambda^k e^{-\lambda}) / k!$$

$$\text{Mean} = \text{Var} = \lambda$$

Normal (Gaussian) Distribution

$$X \sim N(\mu, \sigma^2)$$

$$\text{Mean} = \mu, \text{Var} = \sigma^2$$

68–95–99.7 rule